



Saint Columban College
College of Computing Studies
Pagadian City



Fresh Java

Fresh Java 



Medium

Reverse Engineering

picoCTF 2022

Java

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Description

Can you get the flag?

Reverse engineer this [Java program](#).

Hints 

1

10,884 users solved



75% Liked



picoCTF{FLAG}

Submit Flag

Author: The Analyst: Hyposelenia

Challenge: Fresh Java

Category: Reverse Engineering

Date: 3/31/26



I. Objective

The objective of this challenge is to understand how compiled Java programs can be **reversed (decompiled)** back into readable code.

The goal is to inspect the program's logic and **retrieve the flag directly from the source code.**

II. Background

Java programs are usually written in **human-readable code**, then compiled into .class files which are not easily readable.

However, unlike some compiled languages, Java can often be **decompiled**, meaning:

The compiled file can be converted back into readable code

This makes it possible to:

- analyze program logic
- find hidden strings (like flags)
- understand how the program works

III. Tool Used

- **Online Java Decompiler** - Used to convert the .class file into readable Java source code

IV. Methodology

1. Downloaded the provided Java .class file

- - -
Reverse engineer this **Java program**.



2. Uploaded the file into an **online Java decompiler**

Decompiler.com

JAR String Editor

Drop EXE or DLL,
JAR or CLASS,
APK, XAPK or DEX,
PYC or PYO,
LUAC or LUB,
SWF, ASAR, SMX or AMXX file here
[Choose file](#)

Secure JS file uploader. Easy setup | User Friendly | Reliable. **Start Free.**

3. Decompiled the file into readable Java code

KeygenMe.java

Delete

KeygenMe.java

Download file

```
import java.util.Scanner;

public class KeygenMe {
    public static void main(String[] var0) {
        Scanner var1 = new Scanner(System.in);
        System.out.println("Enter key:");
        String var2 = var1.nextLine();
        if (var2.length() != 34) {
            System.out.println("Invalid key");
        } else if (var2.charAt(33) != '}') {
            System.out.println("Invalid key");
        } else if (var2.charAt(32) != '9') {
            System.out.println("Invalid key");
        } else if (var2.charAt(31) != '8') {
            System.out.println("Invalid key");
        } else if (var2.charAt(30) != 'c') {
            System.out.println("Invalid key");
        } else if (var2.charAt(29) != 'a') {
            System.out.println("Invalid key");
        } else if (var2.charAt(28) != 'c') {
            System.out.println("Invalid key");
        } else if (var2.charAt(27) != '8') {
            System.out.println("Invalid key");
        } else if (var2.charAt(26) != '3') {
            System.out.println("Invalid key");
        }
    }
}
```

4. Manually constructed the flag from the readable code



V. Result

picoCTF{700l1ng_r3qu1r3d_738cac89}

```
picoCTF{700l1ng_r3qu1r3d_738cac89}
```

VI. Explanation

What is Java Code?

Java code is:

Instructions written by a programmer in a readable format

Example:

```
System.out.println("The Analyst: Hyposelenia");
```

This is easy for humans to understand.

What is Compilation?

Before running, Java code is turned into:

.class file

This is:

- machine-readable
- not easy for humans to read



Think of it like:

- Original code = **English sentence**
- Compiled code = **secret coded version**

What is Decompiling?

Decompiling means:

Turning the compiled file back into readable code

VII. Conclusion

This challenge demonstrates how compiled Java programs can be reversed to reveal their original logic and embedded data.

By decompiling the .class file, the flag was easily identified within the source code.

This reinforces an important cybersecurity concept:

Sensitive information should never be hardcoded in programs, as it can be exposed through reverse engineering.

— The Analyst: Hyposelenia